



Tribhuvan University
Faculty of Humanities and Social Sciences

A
PROJECT PROPOSAL
ON
NEPAL BUS TICKET BOOKING SYSTEM
(Course Code: CAPJ356)



Submitted to Department of Information Technology
Kabhre Multiple Campus

*A Proposal submitted for the partial fulfillment of requirements for the degree of
Bachelor in Computer Application (Sixth Semester)*

Submitted by
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October, 2026

Tribhuvan University
Faculty of Humanities and Social Sciences
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Supervisor's Recommendation

I hereby recommend that this project proposal prepared under my supervision by **Nischal Shrestha** entitled “**NEPAL BUS TICKET BOOKING SYSTEM**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the proposal defense.

.....
Signature
[Supervisor Name]
SUPERVISOR
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Letter of Approval

This project proposal entitled “**NEPAL BUS TICKET BOOKING SYSTEM**” by **Nischal Shrestha** as a partial fulfillment of the requirements for the degree of Bachelor in Computer Application (Sixth Semester) has been examined by us and is accepted.

Panel of Examiners:

Name

Signature

Date

1. HOD [HOD Name]

2.

3.

4.

Abstract

Long-distance bus ticketing in Nepal still depends largely on physical counters, phone calls, and local agents. This causes problems for passengers, who cannot easily check seat availability or compare fares, and for operators, who manage seat records manually and face double booking and disputes. This project proposes the Nepal Bus Ticket Booking System, a web-based application that lets passengers search for buses by origin, destination, and travel date, choose seats from a visual seat map, and book tickets online. Bus operators can manage routes, buses, schedules, and fares, and administrators can oversee users, bookings, and reports.

The system prevents double booking through a temporary seat-locking mechanism. It uses sorting and searching algorithms to present results by price or departure time, a fare calculation module that supports special pricing rules, and reports on bookings, revenue, and seat occupancy. Payment is demonstrated through a simulated workflow that can later be replaced by a real payment gateway. The system will be developed as a responsive web application using Next.js, NestJS, PostgreSQL, and Tailwind CSS.

Keywords: bus ticket booking, seat locking, web application, Nepal, NestJS, PostgreSQL

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Road transport is the main means of long-distance travel in Nepal, and buses connect cities such as Kathmandu, Pokhara, Chitwan, and Birgunj to the rest of the country. Despite this, ticket booking for many bus services still depends on physical counters, phone calls, or local agents. Passengers often have to visit the counter in person to check seat availability, compare prices, or confirm a booking, and operators must manage seat records manually.

The Nepal Bus Ticket Booking System is a web-based application that allows passengers to search for buses, view available seats, and reserve tickets online. Passengers can search by origin, destination, and travel date, select seats from a visual seat map, and receive a booking confirmation. Bus operators can manage their routes, buses, schedules, and fares, while administrators oversee users, bookings, and reports.

The system focuses on solving the problem of double booking through a seat-locking mechanism, which holds a seat temporarily while a passenger completes the booking. It uses sorting and searching algorithms to present results by price or departure time, and a fare calculation module for pricing rules. The system will be developed as a responsive, mobile-friendly web application using Next.js, NestJS, PostgreSQL, and Tailwind CSS. Payment will be demonstrated through a simulated payment workflow, designed so a real payment gateway can be added later.

1.2 Problem Statement

Many long-distance bus services in Nepal still rely on manual ticketing. Passengers must visit counters or contact agents to find out which buses are running, whether seats are available, and what the fare is. Comparing the options of different operators on a route takes time, and a passenger often cannot be sure a seat is confirmed until they reach the counter.

Manual booking also creates problems for operators. Seat records are kept on paper or in separate registers, which makes errors likely. When two agents or counters sell the same seat, double booking occurs and leads to disputes at boarding time. Cancellations, fare changes during festival seasons, and sales reports are hard to track without a central system.

Because of these issues, there is a need for a system that shows live seat availability, prevents the same seat from being booked twice, and gives operators one place to manage routes, schedules, fares, and bookings. This project addresses that need by developing a

web-based bus ticket booking system tailored to the Nepali context.

1.3 Objectives

The main objective of this project is to develop a web-based bus ticket booking system that makes seat reservation easier and more reliable for passengers and operators in Nepal.

The specific objectives are:

1. To develop a search module that lets passengers find buses by origin, destination, and travel date, and sort the results by price or departure time.
2. To implement a seat selection and booking module that shows live seat availability and prevents double booking through a temporary seat-locking mechanism.
3. To provide role-based access for passengers, bus operators, and administrators, with secure authentication.
4. To develop a fare calculation module that supports route-based pricing and special pricing rules, such as peak-season fares.
5. To implement a booking and cancellation workflow with a simulated payment step, designed so a real payment gateway can be integrated later.
6. To generate reports for operators and administrators, such as total bookings, revenue, and seat occupancy.

1.4 Scope

The scope of this project covers the design and development of a web-based bus ticket booking system for long-distance routes in Nepal. The system is a responsive web application for desktop and mobile browsers and includes the following:

1. Three user roles (passengers, bus operators, and administrators) with registration, login, secure password storage, and role-based access.
2. Management of routes, buses, schedules, and fares by operators and administrators.
3. Searching for buses by origin, destination, and travel date, with results sorted by price or departure time.
4. A visual seat map showing available and booked seats, with temporary seat locking to prevent double booking.
5. Fare calculation with special rules such as peak-season fares, along with booking, cancellation, and booking history using a simulated payment step.

6. Reports for operators and administrators, such as total bookings, revenue, and seat occupancy.

1.5 Limitations

The system has the following limitations:

1. Payment is simulated. No real money is processed, and integration with a payment gateway such as eSewa or Khalti is left for future work.
2. The system is a prototype that runs on sample data. It is not connected to real bus operators, and the routes and fares are for demonstration only.
3. It does not provide live bus tracking or GPS location.
4. It is a web application only. A native mobile app is not developed.
5. It is not tested for large-scale traffic and is designed for a limited number of users.
6. Notifications such as SMS or email are not included in the current version.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

2.1.1 Study of Existing System

Manual system. In the traditional system, a passenger books a long-distance bus ticket by visiting the operator's counter, calling the office, or going through a local agent. The staff check a paper register or a seat chart, tell the passenger which seats are free, take the payment in cash, and write the booking by hand. Each counter or agent keeps its own record, so seat information is not shared in real time. This leads to double booking, long queues, errors in the records, and difficulty in producing sales reports.

Existing online platforms. Several online platforms already offer bus ticket booking in Nepal. BusSewa describes itself as a real-time booking platform that lets passengers search routes, compare fares, choose seats, and receive e-tickets, with payment through wallets, banks, and cards [1]. GoMyGo allows several passengers to be booked in one transaction, sends confirmations by email and SMS, and follows each operator's own cancellation policy [2]. Okay Journey offers an interactive seat map and a choice of bus class [3]. Trip Turbo lets users filter buses by price and departure time, pick window or aisle seats from a seat map, and receive an e-ticket by email and SMS [4].

Comparison. Table 2.1 compares the manual system, the existing online platforms, and the proposed system.

Summary. Existing platforms have solved the main problems of the manual system for the operators they work with. This project does not aim to replace them. Its purpose is to build the core booking logic as a complete system with its own modules: seat locking to prevent double booking, fare rules, sorting and searching of results, role-based management for passengers, operators, and administrators, and reports. The modules are written for this project and not taken from plugins or third-party services, so they can be studied, tested, and extended.

2.1.2 Key Concepts

[To be written]

2.2 Literature Review

[To be written]

Table 2.1: Comparison of the manual system, existing platforms, and the proposed system

Feature	Manual system	Existing online platforms	Proposed system
Seat availability	Checked at the counter or by phone	Real-time	Real-time
Seat selection	Done by staff, limited	Interactive seat map	Interactive seat map
Booking record	Paper registers	Stored in a database	Stored in a database
Payment	Cash at the counter	Wallets, banks, cards	Simulated (prototype)
Notifications	None	Email and SMS	Not included
Operator reports	Hard to prepare	Offered through the platform	Bookings, revenue, occupancy
Scope	Single counter	Commercial, many operators	Academic prototype, sample data

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